

Kunal Maheshwari

Mount Pleasant, MI, USA | (989) 906-2108 | kunal.maheshwari.work@gmail.com | <https://www.linkedin.com/in/kunalmaheshwari26>

SUMMARY

OPT-EAD holder with over 4 years of experience as a Software Developer and Data Engineer, specializing in building scalable microservices and distributed data pipelines across diverse domains. Proficient in Java, Python, Spark, SQL, and cloud platforms including AWS, Azure, and GCP, with a strong background in CI/CD automation and advanced data analytics. Expertise extends to large-scale data platforms and privacy-preserving AI applications, aiming to leverage robust data querying and relational database skills to drive impactful engineering solutions.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Designed and optimized high-performance REST and SOAP microservices using Java 8/11, Spring Boot, and Azure Functions, consistently handling over 50,000 daily API requests with 99.99 percent SLA uptime.
- Automated CI/CD pipelines with Jenkins, Maven, and Azure DevOps, facilitating continuous delivery across Azure Cloud and Oracle WebLogic environments.
- Spearheaded a significant EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles and improved system stability.
- Modernized legacy systems by migrating to Log4j 2.x, enhancing monitoring and error handling capabilities to reduce issue resolution time by 20 percent.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions, utilizing Git, SVN, and comprehensive JUnit testing for code quality.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Engineered robust Java and EJB-based enterprise applications and RESTful web services for key automotive clients, enhancing client operational efficiency.
- Achieved a 90 percent system performance improvement through meticulous SQL query optimization, database tuning, and modernization of legacy Java code.
- Implemented effective CI/CD pipelines utilizing Jenkins and GitLab, incorporating automated testing and quality gates to ensure software integrity.
- Mentored junior developers by conducting thorough code reviews, leading debugging sessions, and promoting clean coding practices to foster team growth.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems, achieving over 99 percent uptime through proactive monitoring and continuous performance optimization.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and significantly strengthening data integrity.
- Implemented comprehensive unit testing strategies using JUnit4, achieving an impressive 99 percent code coverage across critical modules.
- Supported release planning and sprint execution using JIRA within Agile development teams, ensuring timely project deliveries.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, GCP, AWS, Terraform) January 2024 - May 2025

- Developed a distributed Spark-based data pipeline to process and analyze over 8 GB of NYC taxi trip data in near real time, providing timely insights.
- Engineered an interactive Flask web application for data exploration, featuring dynamic filtering and visualization capabilities for enhanced user interaction.
- Containerized ingestion, processing, and analytics components using Docker for reproducible and scalable deployments across cloud environments.
- Optimized Spark memory usage and partitioning strategies, effectively reducing overall job runtime by 40 percent.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight for efficient resource management.

Privacy-Preserving AI Email Client (Capstone Project - T5, LoRA) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform advanced email summarization and classification tasks.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under stringent privacy constraints.
- Implemented robust model evaluation using ROUGE metrics and integrated quantization for secure and efficient local deployment of AI models.

SKILLS

- Programming Languages: Java, Python, SQL.

- Cloud Platforms: AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), Google Cloud Platform (GCP).
- Databases: PostgreSQL, Oracle Database, IBM DB2, Relational Database management.
- Frameworks & Libraries: Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, LoRA, JUnit.
- Tools & Technologies: Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, Terraform.
- Concepts: ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/CD, Microservices, Large-scale data platforms, Infrastructure automation, Privacy-preserving LLM-based applications.
- Web Services: REST API, SOAP API.
- Data Analysis & Visualization: Tableau, QuickSight, Data Querying.
- Soft Skills: Communication Skills, problem-solving, collaborative teamwork.